Delta Method and Asymptotic Normality of MLEs

Math 742

2/24/2026

Introduction

Introduction

Imagine you are trying to read really small text by using a microscope. The process is not perfect, your eye or hand might wiggle slightly causing you to read a 'j' for an 'i'. In other words, there is noise in your measurement, call this the **estimation error** of $\hat{\theta}_n$. In order to minimize it, you zoom in. At higher zoom levels, the jitter appears **more pronounced**, while at lower levels, it seems **smaller**.

The **delta method** sort of works like that.

- $\hat{\theta}_n$ has some "jitter", the "jitter" can be approximated by:

$$\hat{\theta}_n = \theta + \frac{1}{\sqrt{n}} Z \text{ (where } Z \sim N(0, V))$$

But what happens when we turn the magnification knob? In other words,what if we are actually interested in $g(\theta)$ and not θ. The delta method gives you:

> How the error in $\hat{\theta}_n$ gets affected (magnified or shrunk) by turning the magnification knob (i.e. when θ is transformed by $g(\cdot)$).

When an estimator is **close** to the true parameter value, a smooth transformation can be well-approximated by its **tangent line**. That single idea explains why (i) many estimators look normal in large samples, and (ii) smooth functions of those estimators also look normal.

A useful mental model:

- The estimator has small random "jitter" around the truth;
- Applying a smooth function stretches/shrinks that jitter by the slope of the function at the truth.

Delta Method

Theorem (Delta Method). Suppose

$$\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, V),$$

where $0 < V < \infty$, and g is differentiable at θ with derivative $g'(\theta)$. Then

$$\sqrt{n}\big(g(X_n) - g(\theta)\big) \xrightarrow{d} N\big(0, [g'(\theta)]^2 V\big).$$

Proof sketch (Taylor + Slutsky).

Assume

$$\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, V).$$

$$\boxed{\begin{array}{c} V \text{ is the limiting variance. where } \mathrm{Var}(\sqrt{n}(X_n - \theta)) \to V \\ \text{But, } \mathrm{Var}(\sqrt{n}(X_n - \theta)) = n\mathrm{Var}(X_n) \implies V = \lim_{n\to\infty} n\mathrm{Var}(X_n) \\ \text{So, } \mathrm{Var}(X_n) \approx \frac{V}{n} \text{ for large } n. \end{array}}$$

A first-order Taylor expansion about θ gives

$$g(X_n) = g(\theta) + g'(\theta)(X_n - \theta) + R_n,$$

where the remainder is quadratic in $(X_n - \theta)$ (so R_n is on the order of $(X_n - \theta)^2$).

Since $X_n - \theta$ fluctuates on the order $1/\sqrt{n}$, the remainder is of smaller order (about $1/n$), so $\sqrt{n}R_n \xrightarrow{p} 0$.

Multiplying by $\sqrt{n}$ yields

$$\sqrt{n}\big(g(X_n) - g(\theta)\big) = g'(\theta)\sqrt{n}(X_n - \theta) + \sqrt{n}R_n.$$

By the CLT, the first term converges in distribution to

$$N\big(0, (g'(\theta))^2 V\big),$$

and the second term converges in probability to 0. By Slutsky's theorem (additive version),

$$\sqrt{n}\big(g(X_n) - g(\theta)\big) \xrightarrow{d} N\big(0, (g'(\theta))^2 V\big).$$

(Note: This is the single-parameter version; the multivariate version replaces $g'(\theta)$ with the gradient and V with the covariance matrix.)

Asymptotic Normality of MLEs (quick reminder)

Under standard regularity conditions (support independent of θ, and interchanging differentiating and integration allowed), the MLE satisfies:

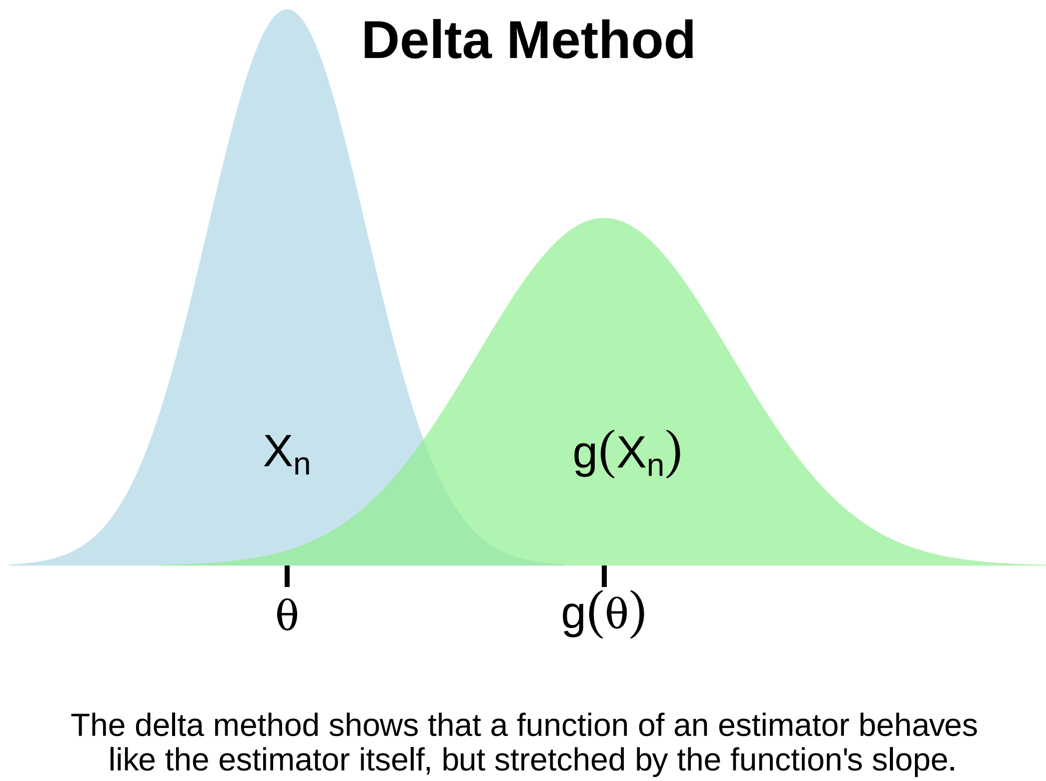
$$\sqrt{n}(\hat{\theta} - \theta) \;\xrightarrow{d}\; N\big(0,\; I(\theta)^{-1}\big)\,,$$

where $I(\theta)$ is the Fisher information (per observation). Equivalently,

$$\hat{\theta} \approx N\left(\theta,\; \frac{1}{n} I(\theta)^{-1}\right) \quad \text{for large } n.$$

This pairs perfectly with the Delta Method: any smooth function $g(\hat{\theta})$ inherits an approximate normal distribution.

Visual intuition: variance "magnification" by the derivative



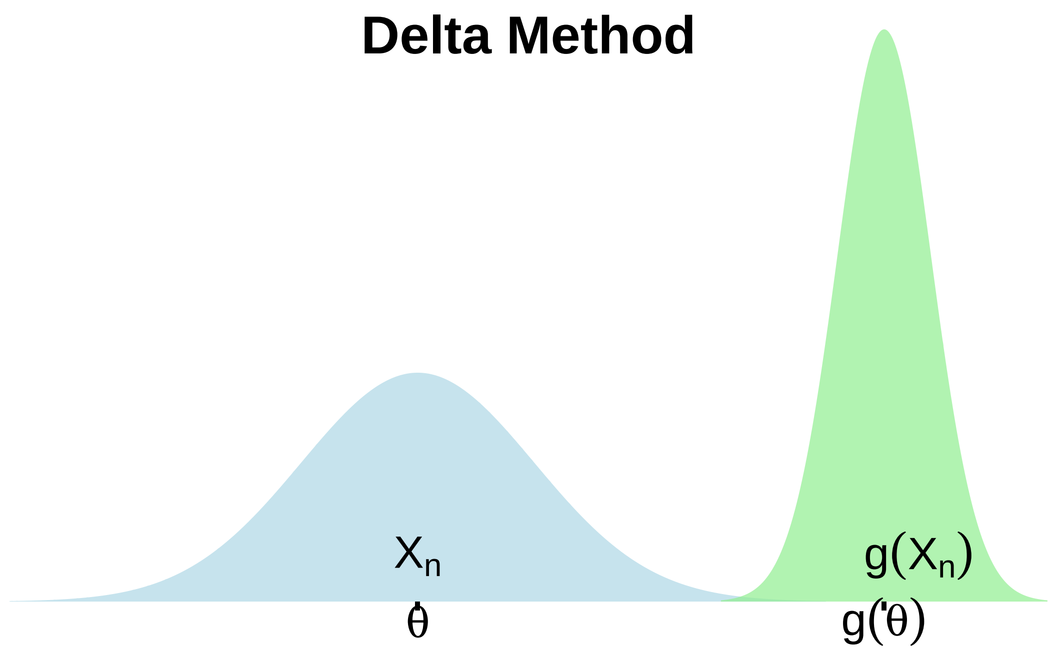
What is this picture showing us? If X_n moves by a tiny amount δ, then $g(X_n)$ moves by approximately $g'(\theta)\delta$. Small deviations of X_n get multiplied by the slope. **The slope is the magnification factor.**

$$\mathrm{Var}(g(X_n)) \approx (g'(\theta))^2 \mathrm{Var}(X_n)$$

- The derivative scales deviations.
- Variance squares deviations.

That's why the derivative gets squared in the Delta Method.

Now, watch what happends if the $g'(\theta) < 1$.



The delta method shows that a function of an estimator behaves like the estimator itself, but squished by the function's slope.

Summary

- The derivative tells how sensitive the output is to the input estimate.
- If g amplifies small changes, variance increases.
- If g shrinks them, variance decreases.

Example 1: Exponential survival median via Delta Method

Suppose $T_1, \ldots, T_n \overset{iid}{\sim} \text{Exponential}(\lambda)$ (rate $\lambda > 0$).

The MLE is

$$\hat{\lambda} = \frac{1}{\bar{T}}.$$

A standard asymptotic result is

$$\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N(0, \lambda^2),$$

so $\text{Var}(\hat{\lambda}) \approx \lambda^2/n$.

The **median survival time** m satisfies $0.5 = 1 - e^{-\lambda m}$, hence

$$m = \frac{\ln 2}{\lambda}.$$

Define $g(\lambda) = \ln(2)/\lambda$. Then

$$g'(\lambda) = -\frac{\ln 2}{\lambda^2}.$$

By the Delta Method,

$$\operatorname{Var}\left(g(\hat{\lambda})\right) \;\approx\; [g'(\lambda)]^2 \;\operatorname{Var}(\hat{\lambda}) = \left(\frac{-\ln 2}{\lambda^2}\right)^2 \frac{\lambda^2}{n} = \frac{(\ln 2)^2}{n\lambda^2}.$$

Because $\hat{\lambda} = 1/\bar{T}$, the approximate standard error is:

$$\mathrm{SE}\left(g(\hat{\lambda})\right) \approx \frac{\ln 2}{\lambda\sqrt{n}} \approx \frac{\ln 2}{\hat{\lambda}\sqrt{n}} = \frac{\ln 2}{\sqrt{n}}\bar{T}.$$

An approximate 95% CI for the median is

$$g(\hat{\lambda}) \;\pm\; 1.96\;\mathrm{SE}\left(g(\hat{\lambda})\right).$$

Simulation check (one plot)

```
n <- 100
B <- 5000
lambda <- 0.5

median_true <- log(2) / lambda

median_hat <- numeric(B)
lambda_hat <- numeric(B)

for (b in 1:B) {
  T <- rexp(n, rate = lambda)
  lam_hat <- 1 / mean(T)
  lambda_hat[b] <- lam_hat
  median_hat[b] <- log(2) / lam_hat
}

emp_sd <- sd(median_hat)
se_delta_plug_in <- mean( log(2) / (lambda_hat * sqrt(n)) )

cat("True median:", median_true, "\n")
```

```
## True median: 1.386294
```

```
cat("Empirical median:", mean(median_hat), "\n")
```

```
## Empirical median: 1.388263
```

```
cat("Empirical SD of median_hat:", emp_sd, "\n")
```

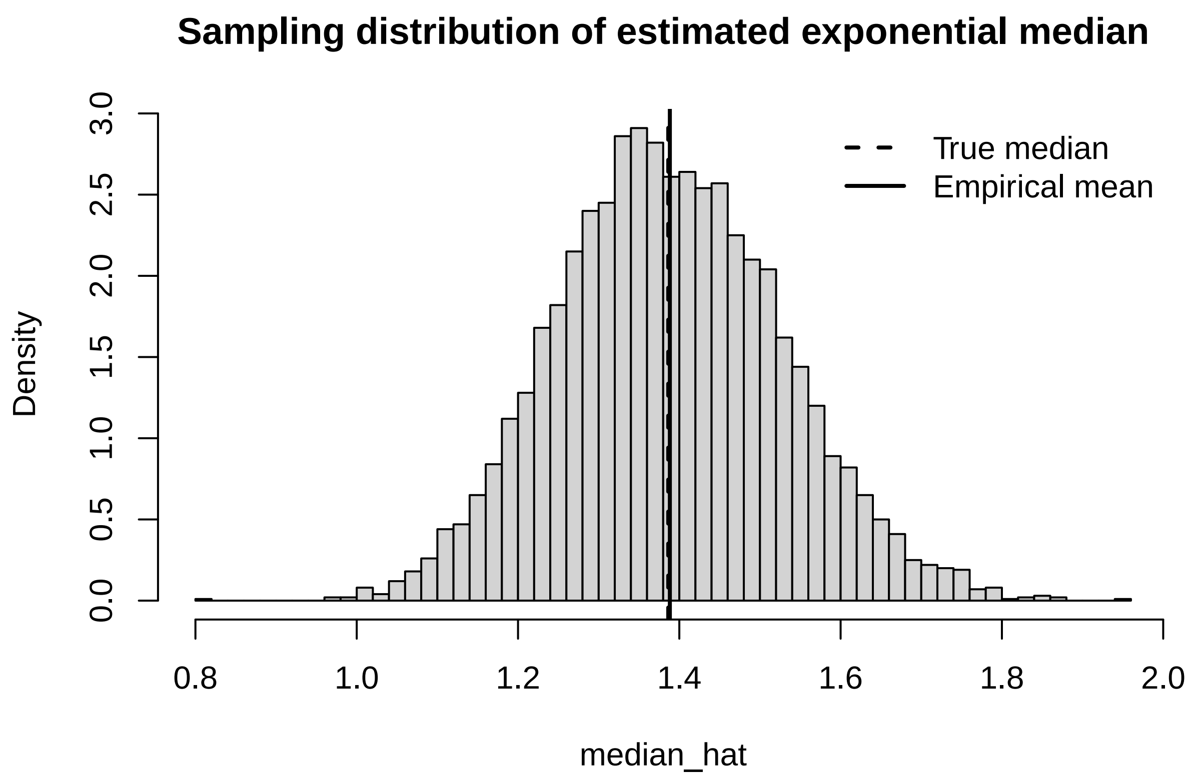
```
## Empirical SD of median_hat: 0.1397527
```

```
cat("Mean plug-in delta SE:", se_delta_plug_in, "\n")
```

```
## Mean plug-in delta SE: 0.1388263
```

```
hist(median_hat, breaks = 50, probability = TRUE,
     main = "Sampling distribution of estimated exponential median",
     xlab = "median_hat")
abline(v = median_true, lwd = 2, lty = 2)
abline(v = mean(median_hat), lwd = 2)
```

```r
legend("topright",
       legend = c("True median", "Empirical mean"),
       lty = c(2,1), lwd = 2, bty = "n")
```



Example 2: Delta Method for $\log(\bar{X})$

Let $X_1, \ldots, X_n$ be i.i.d. with mean $\mu > 0$ and variance σ^2. By the CLT,

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Let $g(x) = \log x$. Then $g'(\mu) = 1/\mu$, and the Delta Method gives

$$\sqrt{n}\big(\log(\bar{X}) - \log(\mu)\big) \xrightarrow{d} N\left(0, \frac{\sigma^2}{\mu^2}\right),$$

so

$$\mathrm{Var}(\log(\bar{X})) \approx \left(\frac{1}{\mu}\right)^2 (\sigma^2/n) = \frac{\sigma^2}{n\mu^2}.$$

Simulation check (one plot)

```r
n <- 200
B <- 5000
mu <- 2
sigma <- 1
```

```r
log_means <- numeric(B)

for (b in 1:B) {
  x <- rnorm(n, mean = mu, sd = sigma)
  log_means[b] <- log(mean(x))
}

emp_sd <- sd(log_means)
se_delta <- (sigma / (sqrt(n) * mu))

cat("True log(mu):", log(mu), "\n")
```

```
## True log(mu): 0.6931472
```

```r
cat("Empirical log(mu):", mean(log_means), "\n")
```

```
## Empirical log(mu): 0.6935455
```

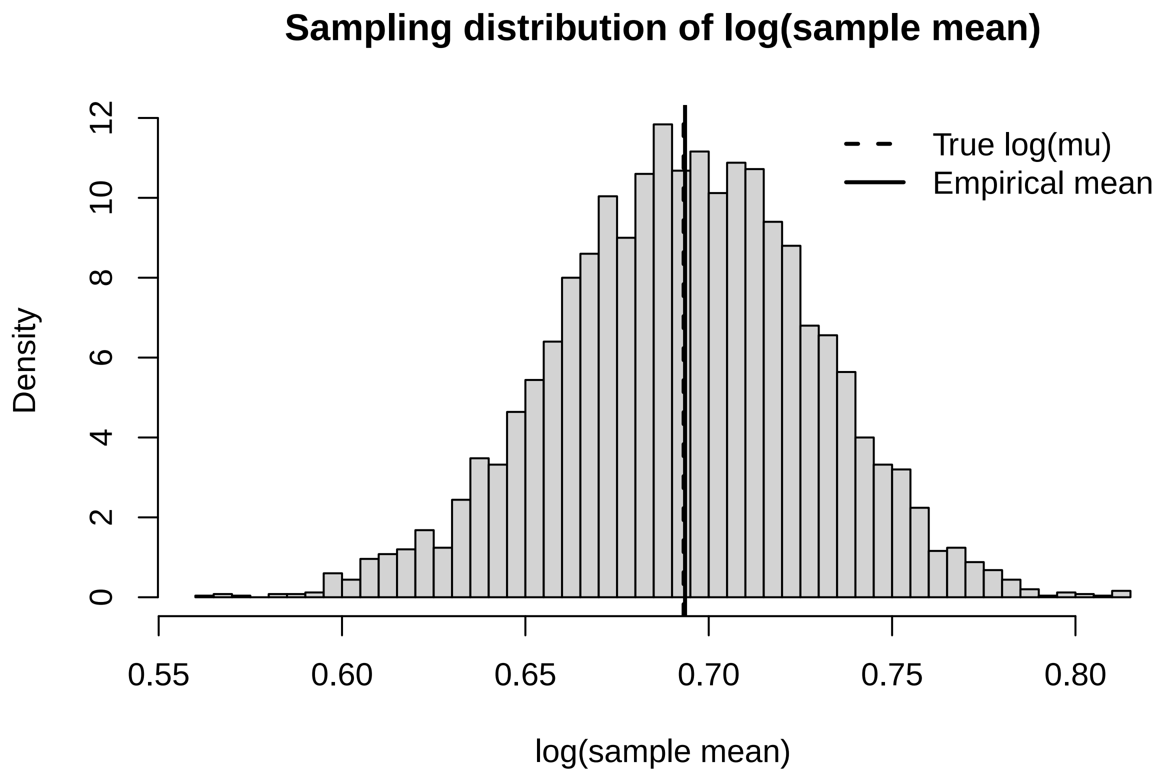
```r
cat("Empirical SD of log(mean):", emp_sd, "\n")
```

```
## Empirical SD of log(mean): 0.03527889
```

```r
cat("Delta-method SE:", se_delta, "\n")
```

```
## Delta-method SE: 0.03535534
```

```r
hist(log_means, breaks = 50, probability = TRUE,
     main = "Sampling distribution of log(sample mean)",
     xlab = "log(sample mean)")
abline(v = log(mu), lwd = 2, lty = 2)
abline(v = mean(log_means), lwd = 2)
legend("topright",
       legend = c("True log(mu)", "Empirical mean"),
       lty = c(2,1), lwd = 2, bty = "n")
```



Summary takeaways

- If $\sqrt{n}(X_n - \theta)$ is asymptotically normal, then $\sqrt{n}(g(X_n) - g(\theta))$ is also asymptotically normal for smooth g.
- The derivative $g'(\theta)$ controls how uncertainty propagates: roughly, $\mathrm{Var}(g(X_n)) \approx [g'(\theta)]^2 \mathrm{Var}(X_n)$.
- Combine **MLE asymptotic normality** with the **Delta Method** to get standard errors and confidence intervals for transformed MLEs.

This is why we love the Delta Method: one derivative gives us standard errors and CIs for any smooth transformation of an MLE.